

(a) Necessity and sufficiency on a shared draw

one shared draw $\mathbf{u} \sim P_{\mathbf{u}}$



shared $\mathbf{u}$:
 $Y^n \perp Y^s \mid \mathbf{u}$

necessity world

do($\mathbf{C} = \mathbf{c}'$), hold $\mathbf{T}$

does removing it change $\mathbf{Y}$?



sufficiency world

do($\mathbf{C} = \mathbf{c}^*$), hold $\mathbf{T}$

does restoring it sustain $\mathbf{Y}$?



$$ci(Y^s, Y^n, y^*)$$

$$\mathbb{E}_{\Gamma \otimes \Delta \otimes P_{\mathbf{u}}}[ci] = \text{PCI score for } X_k$$

OBCB instance

X_k = gender, $\mathbf{T} = \{\text{check-failed}\}$ held at factual,
 $\mathbf{c}^*$ = female (restore), $\mathbf{c}'$ = male (alternative)

(b) Impact kernel: Absolute Difference score (one choice of ci)

